

MA2071

Solutions to Practice midterm

- 1) (i) We have $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ or $ax + by + cz + d = 0$ where $d = -(ax_0 + by_0 + cz_0)$.

(ii) Let $P_0 = (x_0, y_0, z_0) \in P$ and consider the vector $\mathbf{b} = \overrightarrow{P_0 P_1} = \langle x_1 - x_0, y_1 - y_0, z_1 - z_0 \rangle$. Note that D equals the absolute value of the scalar projection of $\mathbf{b}$ onto the normal vector $\mathbf{n} = \langle a, b, c \rangle$. Thus,

$$D = |\text{comp}_{\mathbf{n}} \mathbf{b}| = \frac{|\mathbf{n} \cdot \mathbf{b}|}{|\mathbf{n}|} = \frac{|(ax_1 + by_1 + cz_1) - (ax_0 + by_0 + cz_0)|}{\sqrt{a^2 + b^2 + c^2}}.$$

As $P_0 \in P$, we have $ax_0 + by_0 + cz_0 + d = 0$ and the result follows.

- 2) (i) Let (x, y) approach $(0, 0)$ through the line L given by $y = mx$ where $m \neq 0$. Check that $f(x, y) = f(x, mx) = \frac{m^2 x}{1 + m^4 x^2}$. So, as (x, y) approaches $(0, 0)$ through L , then $f(x, y)$ approaches 0. Now, let (x, y) approach $(0, 0)$ along the parabola P given by $x = y^2$. Check that $f(x, y) = f(y^2, y) = \frac{1}{2}$. So, as (x, y) approaches $(0, 0)$ through P , $f(x, y)$ approaches $\frac{1}{2}$. Thus, the limit does not exist.

(ii) Let $F(x, y, z) = xyz - \cos(x + y + z) = 0$. So, $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{yz + \sin(x + y + z)}{xy + \sin(x + y + z)}$ and $\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{xz + \sin(x + y + z)}{xy + \sin(x + y + z)}$.

- 3) (i) See the notes.

(ii) For $(x, y) \neq (0, 0)$, check that $f_x(x, y) = \frac{x^4 y + 4x^2 y^3 - y^5}{(x^2 + y^2)^2}$ and $f_y(x, y) = \frac{x^5 - 4x^3 y^2 - xy^4}{(x^2 + y^2)^2}$. Using the definition, check that $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$. Also, use the definition to check that $f_{xy}(0, 0) = -1$ and $f_{yx}(0, 0) = 1$. Note that $f_{xy}(x, y)$ is not continuous at $(0, 0)$. To see this, check that $f_{xy} = \frac{x^6 + 9x^4 y^2 - 4x^2 y^4 + 4y^6}{(x^2 + y^2)^3}$. Now, as (x, y) approaches $(0, 0)$ along the x -axis, then $f_{xy}(x, y)$ approaches 1 while as (x, y) approaches $(0, 0)$ along the y -axis, then $f_{xy}(x, y)$ approaches 4. Thus, Clairaut's Theorem does not apply.

(iii) Let $u = x + at$ and $v = x - at$. Then $z = f(u) + g(v)$. So, $\frac{\partial z}{\partial u} = f'(u)$ and $\frac{\partial z}{\partial v} = g'(v)$. Thus, $\frac{\partial z}{\partial t} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial t} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial t} = af'(u) - ag'(v)$ and

$$\frac{\partial^2 z}{\partial t^2} = a \frac{\partial}{\partial t} (f'(u) - g'(v)) = a \left(\frac{df'(u)}{du} \frac{\partial u}{\partial t} - \frac{dg'(v)}{dv} \frac{\partial u}{\partial t} \right) = a^2 f''(u) + a^2 g''(v).$$

Similarly, $\frac{\partial z}{\partial x} = f'(u) + g'(v)$ and $\frac{\partial^2 z}{\partial x^2} = f''(u) + g''(v)$. So, the result follows.

- 4) (i) The direction of fastest change is $\nabla f(x, y) = (2x - 2)\mathbf{i} + (2y - 4)\mathbf{j}$. So, we need to find all points (x, y) where $\nabla f(x, y)$ is parallel to $\mathbf{i} + \mathbf{j}$. This occurs if and only if $(2x - 2)\mathbf{i} + (2y - 4)\mathbf{j} = k(\mathbf{i} + \mathbf{j})$ for some $k \in \mathbb{R}$. Thus, $k = 2x - 2$ and $k = 2y - 4$ and so $y = x + 1$. So, the direction of fastest change is $\mathbf{i} + \mathbf{j}$ at all points on the line $y = x + 1$.

(ii) Let x , y and z be dimensions of the rectangular box. Then $V = xyz$ and $L = \sqrt{x^2 + y^2 + z^2}$. So, $L^2 = x^2 + y^2 + z^2$ and so $z = \sqrt{L^2 - x^2 - y^2}$. This gives $V(x, y) = xy\sqrt{L^2 - x^2 - y^2}$ for $x, y > 0$. Check that

$$V_x = y\sqrt{L^2 - x^2 - y^2} - \frac{x^2 y}{\sqrt{L^2 - x^2 - y^2}} \text{ and } V_y = x\sqrt{L^2 - x^2 - y^2} - \frac{xy^2}{\sqrt{L^2 - x^2 - y^2}}.$$

Now, $V_x = 0$ implies $y(L^2 - x^2 - y^2) = x^2 y$ or $y(L^2 - 2x^2 - y^2) = 0$ and so $2x^2 + y^2 = L^2$ since $y > 0$. Similarly, $V_y = 0$ implies $x(L^2 - x^2 - y^2) = xy^2$ or $x(L^2 - x^2 - 2y^2) = 0$ and so $x^2 + 2y^2 = L^2$ since $x > 0$. Substitute $y^2 = L^2 - 2x^2$ into $x^2 + 2y^2 = L^2$ to get $x^2 + 2L^2 - 4x^2 = L^2$ or $3x^2 = L^2$. Thus, $x = \frac{L}{\sqrt{3}}$ since $x > 0$. This implies $y = \frac{L}{\sqrt{3}}$.

So, the only critical point is $\left(\frac{L}{\sqrt{3}}, \frac{L}{\sqrt{3}}\right)$ which must be the absolute maximum. Thus, the largest possible volume is $V\left(\frac{L}{\sqrt{3}}, \frac{L}{\sqrt{3}}\right) = \frac{L^3}{3\sqrt{3}}$.

- 5) By symmetry, the desired volume V is 8 times the volume V_1 in the first octant.

$$\text{Now, } V_1 = \int_0^r \int_0^{\sqrt{r^2 - y^2}} \sqrt{r^2 - y^2} dx dy = \int_0^r (r^2 - y^2) dy = 2r^3/3. \text{ Thus, } V = 16r^3/3.$$